

Tide retrospective: *Separable-potential factorisation infrastructure*

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Abstract

A single Tide-loop excursion in the `laplace` seabed, exposing the additively-separable factorisation pattern that the existing `SemiDegenerate.lean` (harmonic + quartic) and `PureQuartic.lean` (quartic + quartic) both re-derive case by case. The new module¹ adds 156 lines of Lean to the seabed, with five generic theorems: a definition, the Boltzmann decomposition, partition-function factorisation, separable-observable factorisation, separable-Gibbs-expectation factorisation, and the mixed-covariance vanishing $\text{Cov}[f \otimes 1, 1 \otimes g] = 0$. Zero `sorry`, zero `axiom`, zero `native_decide`; clean diagnostics on first compile. The interesting moves were upstream: GPT-5.5 Pro flagged a wrong covariance formula in the candidate set (the cross-mean terms $(a_1 b_2 + a_2 b_1) \cdot \langle x \rangle_U \cdot \langle y \rangle_V$ actually cancel under separable Gibbs, never appear in the final covariance), and recommended a tighter set of hypotheses on the partition-factorisation that the Bochner unconditional form would not have caught.

1 Setting

This is the first “infrastructure abstraction” Tide step in the `laplace` seabed: it does not extend the formalised territory outward by adding a new headline theorem, but factors the existing shoreline by extracting a shared template from two parallel files. Both `SemiDegenerate.lean` (harmonic + quartic, 514 lines) and `PureQuartic.lean` (quartic + quartic, 478 lines) carry a structurally identical proof skeleton: define a 2D potential as a sum, prove the Boltzmann factor decomposes as a product, apply Mathlib’s `integral_prod_mul` to factor the partition function, then expand to closed form via the 1D pieces.

The Tide step abstracts that template once for any $L(x, y) = U(x) + V(y)$. Future 2D potentials (notably the quartic–sextic mixed and any subsequent monomial–monomial pair) specialise rather than copy-paste-rewrite. The deferred-but-cost-positive refactor was identified at the end of the 2D pure-quartic retrospective² as the next abstraction worth lifting; the 6 May candidates survey lifted it to candidate A3 and recommended it as the high-yield refactor pick of the post-Tide-10 batch. The user picked A3 directly from the survey digest.

The seabed was `main` at `1e3802a` (the merge of Tide 11, `harmonic-gibbsregularity`); the new branch is `tide/separable-potential`, off `main` (no chain needed this time — all prior tides are merged).

¹Module: `Laplace.TwoD.AddSeparable`.

²`lean/laplace/retrospectives/2026-05-02-tide-2d-pure-quartic.tex`, §Lessons.

2 The thing formalised

For any pair of potentials $U, V : \mathbb{R} \rightarrow \mathbb{R}$, the additively-separable 2D potential is $L(x, y) = U(x) + V(y)$. The module exposes five theorems, of which the headline two are:

Under integrability of each marginal Boltzmann weight,

$$Z_{2D}(t) = Z_U(t) \cdot Z_V(t)$$

where $Z_{2D}(t) = \int e^{-tL(x,y)} dx dy$ and $Z_U(t) = \int e^{-tU(x)} dx$, $Z_V(t) = \int e^{-tV(y)} dy$. For separable observables $\varphi(x, y) = f(x)$, $\psi(x, y) = g(y)$, when both 1D partition functions are nonzero,

$$\text{Cov}_{2D,t}[\varphi, \psi] = 0.$$

The Lean signatures (abbreviated; full file at `Laplace/TwoD/AddSeparable.lean`):

```
noncomputable def addSeparable (U V : ℝ → ℝ) : ℝ × ℝ → ℝ :=
  fun z   U z.1 + V z.2

theorem partitionFunction_addSeparable_factor
  {U V : ℝ → ℝ} {t : ℝ}
  (hU : Integrable (fun x   Real.exp (-(t * U x))))
  (hV : Integrable (fun y   Real.exp (-(t * V y)))) :
  partitionFunction (addSeparable U V) t =
    Laplace.partitionFunction U t * Laplace.partitionFunction V t

theorem integral_separable_addSeparable
  {U V : ℝ → ℝ} {t : ℝ} (f g : ℝ → ℝ)
  (hf : Integrable (fun x   f x * Real.exp (-(t * U x))))
  (hg : Integrable (fun y   g y * Real.exp (-(t * V y)))) :
  ( z : ℝ × ℝ, f z.1 * g z.2 * Real.exp (-(t * addSeparable U V z))) =
    ( x, f x * Real.exp (-(t * U x))) * ( y, g y * Real.exp (-(t * V y)))

theorem gibbsCov_addSeparable_fst_snd_eq_zero
  {U V : ℝ → ℝ} {t : ℝ} (f g : ℝ → ℝ)
  (hZU_ne : Laplace.partitionFunction U t ≠ 0)
  (hZV_ne : Laplace.partitionFunction V t ≠ 0)
  (hU : Integrable ...) (hV : Integrable ...)
  (hf : Integrable ...) (hg : Integrable ...) :
  gibbsCov (addSeparable U V) t (fun z   f z.1) (fun z   g z.2) = 0
```

The mixed-covariance vanishing is the structural payoff of separability: under a product Gibbs measure, the covariance between an observable that depends only on the first coordinate and one that depends only on the second is identically zero. This is the lemma that the existing `cov_affine_*` theorems implicitly use when they observe that the $a_1 b_2$ and $a_2 b_1$ cross terms drop out.

Numerical sanity: not feasible (structural identities, not specific numerical values). The seabed's existing 2D files have already validated the concrete instances numerically.

3 Proof strategy

All four non-trivial theorems pivot on a single Mathlib lemma:

$$\int_{\mathbb{R}^2} f(z_1) g(z_2) d(\mu \times \nu) = \left(\int f d\mu \right) \cdot \left(\int g d\nu \right).$$

This is `MeasureTheory.integral_prod_mul` in Mathlib, and it is unconditional: when either factor is non-integrable the equation holds with both sides equal to 0 by Bochner convention. After the Boltzmann decomposition $e^{-t(U(z_1)+V(z_2))} = e^{-tU(z_1)} \cdot e^{-tV(z_2)}$ via `Real.exp_add`, all of partition, integral-of-separable-observable, and Gibbs-expectation factorisations are one application of `integral_prod_mul` away.

The mixed-covariance corollary is the only real proof in the file: $\text{Cov}[\varphi, \psi] = \langle \varphi \psi \rangle - \langle \varphi \rangle \langle \psi \rangle$. The first term factors via the expectation lemma at (f, g) . The second term factors as $\langle f \rangle_U \cdot 1 \cdot 1 \cdot \langle g \rangle_V$, after applying the expectation lemma to each of $\varphi = f \otimes 1$ and $\psi = 1 \otimes g$ separately and using `Laplace.gibbsExpectation_const` to collapse the constant factors. Then the two products are equal and `ring` closes.

The proof of the constant-factor collapse is a four-line β -eta gymnastics: the helper expects an observable of the form $f(z) = h(z_1) \cdot k(z_2)$, and we want to apply it at $k = 1$ to get $\varphi = f \otimes 1 = f(z_1) \cdot 1$, then rewrite $f(z_1) \cdot 1 = f(z_1)$. The `simp` using `hV` pattern handles the integrability witness for $1 \cdot e^{-tV(y)}$ via `one_mul`.

4 Roadblocks and resolutions

The deliberation surfaced two issues; the formalisation surfaced one.

GPT correction on Candidate B’s covariance formula. My v1 statement of the affine-covariance template (Candidate B) had:

$$\text{Cov}_t[a_1x + a_2y + c, b_1x + b_2y + d] = a_1b_1\text{Var}_U + a_2b_2\text{Var}_V + (a_1b_2 + a_2b_1)\langle x \rangle_U \langle y \rangle_V.$$

GPT pointed out that under a separable Gibbs law, the cross terms $\langle x \rangle_U \langle y \rangle_V$ *cancel* between the joint expectation $\langle (a_1x + a_2y + c)(b_1x + b_2y + d) \rangle$ and the product of marginals $\langle a_1x + a_2y + c \rangle \cdot \langle b_1x + b_2y + d \rangle$. The clean formula is just

$$\text{Cov}_t[\varphi, \psi] = a_1b_1 \cdot \text{Cov}_U[x, x] + a_2b_2 \cdot \text{Cov}_V[y, y].$$

This wasn’t load-bearing for the present tide (we deferred B), but it matters for the follow-up tide that picks up the affine template. The existing files’ `cov_affine_*` theorems do the cancellation inline in their algebraic-manipulation step; the abstracted version states the clean formula directly.

Hypothesis form on ‘`partitionFunction_addSeparable_factor`’. My v1 had the partition factorisation *unconditional*, relying on the Bochner convention that $\int = 0$ on non-integrable functions makes the equation true even without integrability. GPT recommended adding integrability hypotheses anyway, on the grounds that downstream files always have integrability witnesses available and the hypothesis-bearing form is the one the reader naturally chains through:

GPT-5.5 Pro: For Bochner $\int$, unconditional factorisation statements are brittle because the theorem you will actually use is typically the integrable version. ... Downstream files will naturally be able to apply [the hypothesis-bearing form], since for concrete potentials they already prove finiteness/integrability of the partition function.

The committed file takes the integrability hypotheses but doesn't use them in the proof body, since `integral_prod_mul` is unconditional. The hypotheses are documentation / API discipline rather than load-bearing. This is unusual but defensible: a future Mathlib refactor that adds integrability hypotheses to `integral_prod_mul` would not break our callers.

Concurrent-agent branch race (recurrence). A parallel session created its own tide branch off `main` while my work was in flight, and `git commit` fired against *that* branch's checkout instead of the intended `tide/separable-potential`. The fix was a two-line cleanup³. This is the third such recurrence (2D pure-quartic, sextic-moments, now separable-potential). The structural fix — run each Tide agent in its own `git worktree` — remains a recommended skill change rather than landed infrastructure. (After the cleanup, this Tide moved to a worktree at `lean/laplace-separable/` for the remainder of the run; the parallel-session contention disappeared immediately.)

Beyond these three, no proof-side roadblocks. All five theorems typechecked clean on the first compile.

5 What was Mathlib, what was new

Mathlib provided. `MeasureTheory.integral_prod_mul` was the load-bearing lemma; the four non-trivial theorems all reduce to one application of it after a `Real.exp_add` for the Boltzmann decomposition. The 1D Gibbs framework (`Laplace.partitionFunction`, `Laplace.gibbsExpectation`, `Laplace.gibbsCov`, `Laplace.gibbsExpectation_const`) came from the seabed's own `Laplace.Gibbs` module. The 2D Gibbs framework (the `Laplace.TwoD.-` prefixed wrappers used in `SemiDegenerate.lean`) came from the same.

What was new in this tide. The five theorems in the new module: the additively-separable potential definition, its Boltzmann decomposition lemma, partition-function factorisation, separable-observable integral factorisation, Gibbs-expectation-of-separable factorisation, and the mixed-covariance vanishing. None of the five theorems introduces new mathematical content; the value is in stating the abstraction once with the right hypothesis package, where “the right” was determined by GPT-5.5 Pro's critique of the v1 hypotheses.

6 Lessons

- *The deliberation step pays for itself when the candidate is structural.* For algebraic / asymptotic Tides (`kappa3-harmonic`, `anharmonic-kappa3`) GPT's contribution is usually a strategic move inside the proof. For an abstraction Tide like this one, GPT's contribution is on the *statement* — specifically, the hypothesis package and the algebraic cancellation that hides in the naive form. Both interventions saved follow-up cycles in the formalisation.
- *Unconditional factorisation theorems are an anti-pattern even when valid.* `integral_prod_mul` is unconditional in Mathlib, but stating our partition factorisation unconditionally would have invited downstream confusion. Take integrability hypotheses anyway, even if the proof body doesn't use them; they document the API and survive future Mathlib tightening.

³`git branch -f tide/separable-potential <SHA>` to relocate the commit, then `git branch -f <other-tide> main` to restore the parallel session's branch baseline.

- *Concurrent Tide agents continue to need filesystem isolation.* Three tides have now been bitten by the shared-checkout branch race. This recurrence makes the case for landing `git worktree` support in the Tide skill’s preflight check rather than continuing to hand-craft the recovery dance.

7 Follow-ups

- *Refactor the existing 2D files to use the abstraction.* Both `SemiDegenerate.lean` and `PureQuartic.lean` carry `partitionFunction_factor` and `integral_pow_pow_factor` lemmas that are now corollaries of the new module’s theorems at concrete (U, V) choices. The follow-up tide replaces ~ 30 lines per file with ~ 5 -line specialisations.
- *Generic Gibbs-covariance algebra.* GPT recommended a small covariance API in the base Gibbs files: `gibbsCov_symm`, `gibbsCov_add_left/right`, `gibbsCov_const_left/right`, `gibbsCov_smul_left/right`. Once those land, the affine covariance template is short and pleasant, and Candidate B from this deliberation becomes essentially trivial.
- *Mixed-power 2D moments (A4 from the 6 May survey).* Now a one-line corollary of `integral_separable_addSeparable` at $f = x \mapsto x^m$, $g = y \mapsto y^n$. Pairs naturally with the “even-monomial template” tide (A1).
- *Quartic–sextic mixed potential (A2 from the 6 May survey).* Now a specialisation of the new abstraction with closed-form $U = \text{quartic}$ and $V = \text{sextic}$. The deferred 2D Tide.
- *Affine-covariance template (Candidate B / future tide).* Picking up the deferred piece, ideally after the generic Gibbs-covariance algebra lands. The clean statement (no surviving cross-mean terms) is in this retrospective for the future tide to consume.

Acknowledgements

GPT-5.5 Pro contributed both deliberation-stage corrections that shaped the final theorem statements: the Bochner-integrability hypothesis discipline and the algebraic cancellation in the affine covariance formula. The full consult and the tide log are preserved in the primer project’s tide-log directory under the SRI repo.